

Primary sources & further reading

Part 07 —  Agentic AI Systems · 07-agentic-ai/papers/PAPERS.md · 07-agentic-ai/resources/RESOURCES.md

Papers & Primary Sources — Agentic AI Systems

Read the first two before anything else — they're not academic papers but they are the two most consequential primary sources in this space, and they disagree with each other in an instructive way.

Read these two first (and hold them against each other)

#	Source	Why it matters	Link
1	How we built our multi-agent research system — Anthropic Engineering	The primary source for the 15x token-cost figure used throughout this module, and an honest account of when a lead-agent/sub-agent supervisor topology earned its cost. Read the section on why multi-agent won for open-ended research and lost for narrower tasks.	https://www.anthropic.com/engineering/multi-agent-research-system
2	Don't Build Multi-Agents — Walden Yan, Cognition	The sharpest published counter-argument: context isolation is oversold, hand-off summarization loses information structurally, and most tasks are better served by one long-context agent. Read this right after #1 — the two sources are responding to the same trade-off and reaching different defaults for different task shapes.	https://cognition.ai/blog/dont-build-multi-agents

Multi-agent frameworks and architectures (primary sources)

Paper	Year	One-line takeaway	Link
AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation — Wu et al.	2023	Conversable agents as the unit of composition; customizable roles, human-in-the-loop as a first-class agent type	arXiv:2308.08155
MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework — Hong et al.	2023	Encodes Standard Operating Procedures into agent roles to fight "cascading hallucination" from naively chaining LLMs — the pipeline topology, formalized for software engineering	arXiv:2308.00352
CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society — Li et al.	2023	Role-playing agents that negotiate a task via structured dialogue with minimal human steering — an early, clean statement of peer-to-peer coordination	arXiv:2303.17760
Generative Agents: Interactive Simulacra of Human Behavior — Park et al.	2023	Memory, reflection, and planning loops for agents in a shared simulated environment; the ancestor of most "agent society" and long-horizon memory designs	arXiv:2304.03442
Improving Factuality and Reasoning in Language Models through Multiagent Debate — Du et al.	2023	The empirical case for the debate/vote topology: independent models arguing over multiple rounds measurably reduce factual errors versus a single model	arXiv:2305.14325

Evaluation (connects to Module 15)

Paper	Year	One-line takeaway	Link
AgentBench: Evaluating LLMs as Agents — Liu et al.	2023	Multi-environment benchmark exposing how much of "agent" performance is really long-horizon reasoning and instruction-following, not single-turn capability — read this before trusting any single-number agent leaderboard	arXiv:2308.03688

Standards and protocols

Source	What it is	Link
Announcing the Agent2Agent Protocol (A2A) — Google Developers Blog	The original announcement: Agent Cards, Tasks, and the transport for agent-to-agent delegation across vendors	https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/
A2A Protocol Specification	The living technical spec, now under Linux Foundation stewardship	https://a2a-protocol.org/latest/specification/
Model Context Protocol	The agent-to-tool/data standard this module contrasts A2A against — full coverage in Module 09 [09-mcp]	https://modelcontextprotocol.io/
LangGraph — Multi-agent concepts	How this module's five topologies compile down to subgraphs and hand-offs in one popular framework	https://langchain-ai.github.io/langgraph/concepts/multi_agent/

How to read an agentic-systems paper (30 minutes, 3 passes)

1. **Pass 1 (5 min):** what's the topology (even if the paper doesn't name it that way), and what's the claimed win over a single agent?
2. **Pass 2 (15 min):** look for the cost accounting. Does the paper report token/API-call counts against its single-agent baseline, or only task success? A paper that only reports success without cost is telling you half the trade-off from notes/02 section 1.
3. **Pass 3 (10 min):** find the failure analysis. Almost every multi-agent paper has a "limitations" paragraph describing one of the five coordination failure modes in notes/02 section 3 without naming it that way — that paragraph is usually the most useful one in the paper.

Keep a one-paragraph note per paper, specifically on which of the five topologies (notes/01 section 2) it actually implements — paper authors rarely use this module's vocabulary, and mapping their architecture onto it is the fastest way to tell if a "novel framework" is actually a supervisor with new branding.

Resources — Agentic AI Systems

Courses & talks

Resource	Why	Link
Anthropic — Building Effective Agents	The essay this module's "one agent with better tools" position is drawn from	https://www.anthropic.com/research/building-effective-agents
Anthropic — How we built our multi-agent research system	The 15x-token-cost finding, straight from the team that shipped it	https://www.anthropic.com/engineering/multi-agent-research-system
Cognition — Don't Build Multi-Agents	The dissenting, equally-informed view — read both and form your own position	https://cognition.ai/blog/dont-build-multi-agents
DeepLearning.AI — Multi AI Agent Systems with crewAI	Free short course, topology patterns in a framework-specific setting	https://www.deeplearning.ai/short-courses/multi-ai-agent-systems-with-crewai/
Berkeley CS294 — Large Language Model Agents	Academic treatment of planning, tool use, and multi-agent coordination	https://lmsagents-learning.org/

Repositories worth reading

Repo	What's inside
https://github.com/langchain-ai/langgraph	The topology primitives this module describes in the abstract — see ../16-ai-tech-stack/tracks/langgraph/
https://github.com/langchain-ai/langgraph-supervisor-py	A thin, readable reference implementation of the supervisor pattern
https://github.com/langchain-ai/langgraph-swarm-py	A thin, readable reference implementation of the hand-off/swarm pattern
https://github.com/microsoft/autogen	A different design point for multi-agent conversation patterns
https://github.com/crewAIInc/crewAI	Role-based multi-agent framework — a good second reference topology
https://github.com/openai/swarm	OpenAI's minimal, educational multi-agent hand-off primitives
https://a2a-protocol.org/	The Agent2Agent protocol spec referenced in notes/02 §2

Communities

- r/LocalLLaMA and r/MachineLearning — active discussion of agent architecture trade-offs
- LangChain Discord / Slack — practical topology debugging war stories
- Anthropic's Discord — agent design discussions specific to Claude-based systems

Where to go next

- ../06-ai-agents/ if you arrived here first — this module assumes single-agent theory.
- ../16-ai-tech-stack/tracks/langgraph/ for the runtime that implements every topology in notes/01 §2.
- ../15-ai-evals/ for how to actually measure whether a multi-agent system is worth its cost.